

Ex No: 3b

IMPLEMENTATION OF A FUZZY INFERENCE SYSTEM

Aim:

To implement Fuzzy Inference System.

Procedure:

1. Define the fuzzy input variable as temperature and output variable as fan speed.
2. Define Low, Medium and High membership functions using triangular membership functions.
3. Plot and visualize the temperature and fan speed membership functions.
4. Give a crisp temperature input and perform fuzzification.
5. Apply the fuzzy rules relating temperature to fan speed.
6. Combine the activated output membership functions and perform centroid defuzzification.
7. Display the temperature and the resulting crisp fan speed.

Program:

Example: Simple Fuzzy Inference System for Temperature Control

```
import numpy as np
import skfuzzy as fuzz
import matplotlib.pyplot as plt

# Step 1: Define the fuzzy variables (input: temperature, output: fan speed)
temperature = np.arange(0, 41, 1) # Temperature from 0 to 40 degrees Celsius
fan_speed = np.arange(0, 11, 1) # Fan speed from 0 to 10

# Step 2: Define fuzzy membership functions
# Temperature membership functions: Low, Medium, High
temp_low = fuzz.trimf(temperature, [0, 0, 20])
temp_medium = fuzz.trimf(temperature, [10, 20, 30])
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temp_high = fuzz.trimf(temperature, [20, 30, 40])

# Fan speed membership functions: Low, Medium, High
fan_low = fuzz.trimf(fan_speed, [0, 0, 5])
fan_medium = fuzz.trimf(fan_speed, [2, 5, 8])
fan_high = fuzz.trimf(fan_speed, [5, 10, 10])

# Step 3: Visualize the membership functions
plt.figure(figsize=(10, 6))

plt.subplot(2, 1, 1)
plt.plot(temperature, temp_low, label='Low')
plt.plot(temperature, temp_medium, label='Medium')
plt.plot(temperature, temp_high, label='High')
plt.title("Temperature Membership Functions")
plt.xlabel("Temperature (°C)")
plt.ylabel("Membership Degree")
plt.legend()

plt.subplot(2, 1, 2)
plt.plot(fan_speed, fan_low, label='Low')
plt.plot(fan_speed, fan_medium, label='Medium')
plt.plot(fan_speed, fan_high, label='High')
plt.title("Fan Speed Membership Functions")
plt.xlabel("Fan Speed")
plt.ylabel("Membership Degree")
plt.legend()

plt.tight_layout()
plt.show()

```

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# Step 4: Fuzzification - Define a crisp temperature input
temperature_input = 28 # Example temperature

# Fuzzify the input
temp_low_level = fuzz.interp_membership(temperature, temp_low, temperature_input)
temp_medium_level = fuzz.interp_membership(temperature, temp_medium,
temperature_input)
temp_high_level = fuzz.interp_membership(temperature, temp_high, temperature_input)

# Step 5: Apply fuzzy rules
# Rule 1: If temperature is Low, fan speed is Low
# Rule 2: If temperature is Medium, fan speed is Medium
# Rule 3: If temperature is High, fan speed is High

fan_low_level = temp_low_level
fan_medium_level = temp_medium_level
fan_high_level = temp_high_level

# Step 6: Defuzzification - Calculate the crisp output (fan speed)
fan_output = fuzz.defuzz(fan_speed, fan_low * fan_low_level + fan_medium *
fan_medium_level + fan_high * fan_high_level, 'centroid')

# Step 7: Print the output
print(f"Temperature: {temperature_input}°C")
print(f"Fuzzified fan speed: {fan_output:.2f}")

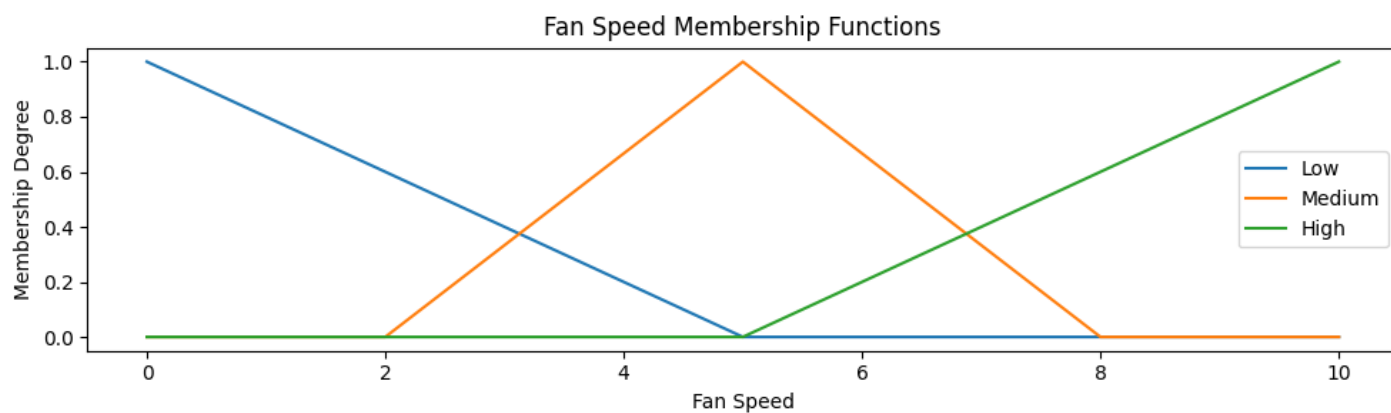
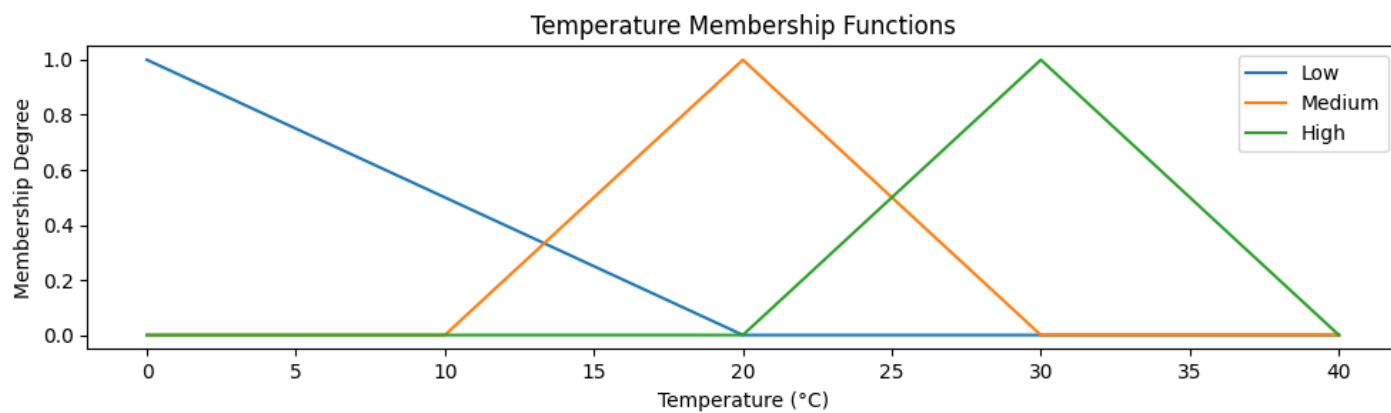
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Output

```

Temperature: 28°C
Fuzzified fan speed: 6.00

```



Temperature: 28°C
Fuzzified fan speed: 7.56